GEM521_Lab4: Individual tree detection using lidR

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```r
library(tidyverse)
library(terra)
library(lidR)
library(rgl)
```

```r
norm_dir <- "Normalized"
plots_csv <- file.path("Plots/Lab4_Plots.csv")

out_dir   <- file.path("Outputs")
plots_dir <- file.path("Plots")           # where you'll save extracted plots
dir.create(out_dir,   showWarnings = FALSE, recursive = TRUE)
dir.create(plots_dir, showWarnings = FALSE, recursive = TRUE)
```

Question 1

In what type of forest stand do Li et al., 2012 suggest their algorithm is successful? Do the four plots you extracted match this description?

- Mixed and complex coniferous forests.
- Yes

Question 2

Which part of the tree do Li et al., 2012 begin their point classification with? Why do they do this?

They begin their classsification from the top of the tree. This is because the spacing between trees is higher at the top, so differentiation may be quite easier than at the lower storey where branches of trees may be easily confused with those of nearby trees.

Question 3

Define the speed_up argument in the `li2012` algorithm.

The speed_up argument is a numeric argument (default = 10) that represents the maximum radius of a crown. The lower the value, the faster the algorithm. Higher values increase the search radius, which slows the algorithm down.

Point Extraction

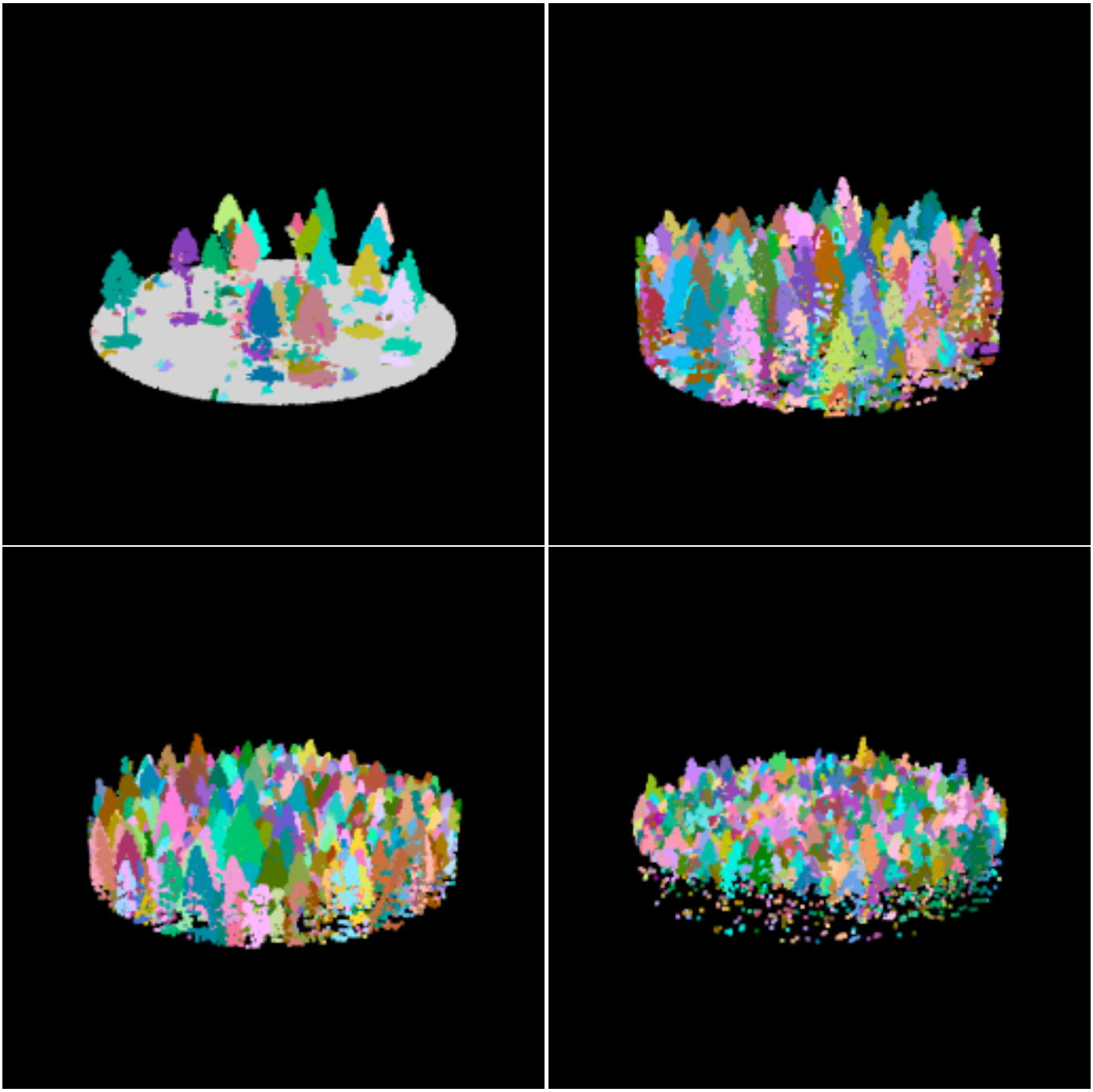
```
# Read normalized LAS catalog
ctg <- readLAScatalog(norm_dir)

# Drop height outliers (same idea as Lab 2)
opt_filter(ctg) <- "-drop_z_below 0 -drop_z_above 65"
opt_progress(ctg) <- FALSE

plots_tbl <- read_csv(plots_csv, show_col_types = FALSE)

radius <- 77 # 154 m diameter(77 m radius)

for(i in 1:4) { #extract plots 1 - 4
  x <- plots_tbl$X[i]
  y <- plots_tbl$Y[i]
  p <- clip_circle(ctg, x, y, radius)
  writeLAS(p, file.path(plots_dir, paste0("Lab4_Plot_", i, ".las")))
}
```

```
## Warning: package 'future' was built under R version 4.5.1
```

Screenshot 1: Plot 1 - 4 Segmentation with li2012 colored by Tree ID

```
li12 <- li2012(dt1 = 1.5, dt2 = 2, R = 2, Zu = 15, speed_up = 10)

for(i in 1:4){

  Plt <- readLAS(file.path("Plots", paste0("Lab4_Plot_", i, ".las")))
  Plt_li12 <- segment_trees(Plt, li12)

  rgl::close3d()
  rgl::open3d(useNULL = TRUE)  # safe for knitting

  plot(Plt_li12, color = "treeID")
```

```
  rgl::rgl.snapshot(file.path("Outputs",
                  paste0("Plot", i, "_Li2012.png")))

  rgl::close3d()
}
```



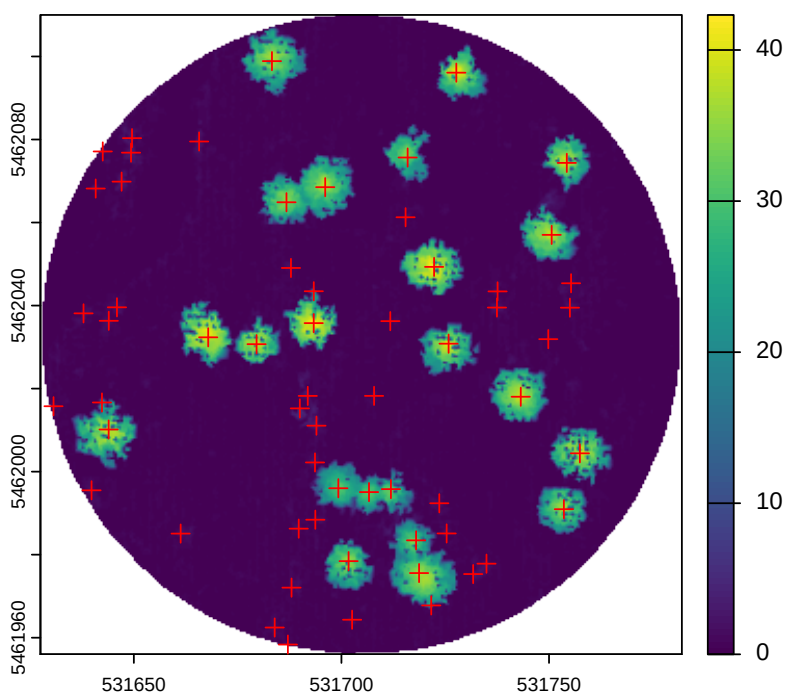
- **Create a CHM for Plot 1**

```r
p1 <- readLAS("Plots/Lab4_Plot_1.las")
```

```r
chm <- rasterize_canopy(
  p1,
  res = 0.5,
  algorithm = pitfree(
    thresholds = c(0,10,20,30),
    subcircle = 0.2,
    max_edge = c(0,1)
  )
)
```

- **Locate Tree Tops**

```
## Warning in plot.window(...): "zebra" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "zebra" is not a graphical parameter

## Warning in title(...): "zebra" is not a graphical parameter

## Warning in plot.sf(ttops, add = TRUE, col = "red", pch = 3): ignoring all but
## the first attribute
```



Question 4

What is a key advantage of the tree-centric approach over existing area-based methods according to Dalponte and Coomes 2016?

The tree-centric method provides more detailed structural information and is better suited for studies focused on individual tree metrics such as crown area and height. In contrast, area-based methods only summarizes canopy characteristics over grid cells without isolating individual trees.

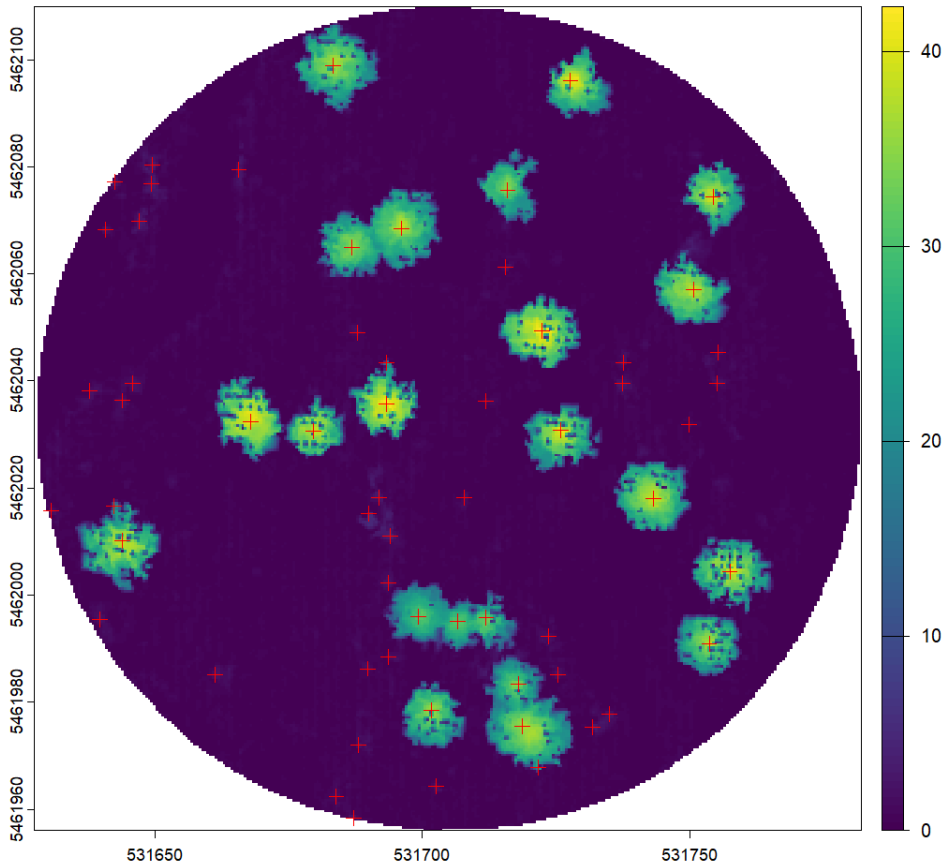
Question 5

Explain the difference between the `th_seed` and `th_cr` arguments of the `dalponte2016` algorithm.

`th_seed` controls the minimum height difference required for a pixel to qualify as a tree seed (local maximum), and it ensures that only the canopy peaks are selected as tree tops.
`th_cr` controls the crown region growing threshold, and determes how far neighboring pixels can extend from the seed

Tree Segmentation

```
## Warning in plot.window(...): "zebra" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "zebra" is not a graphical parameter

## Warning in title(...): "zebra" is not a graphical parameter

## Warning in plot.sf(ttops, add = TRUE, col = "red", pch = 3): ignoring all but
## the first attribute

## pdf
##   2
```



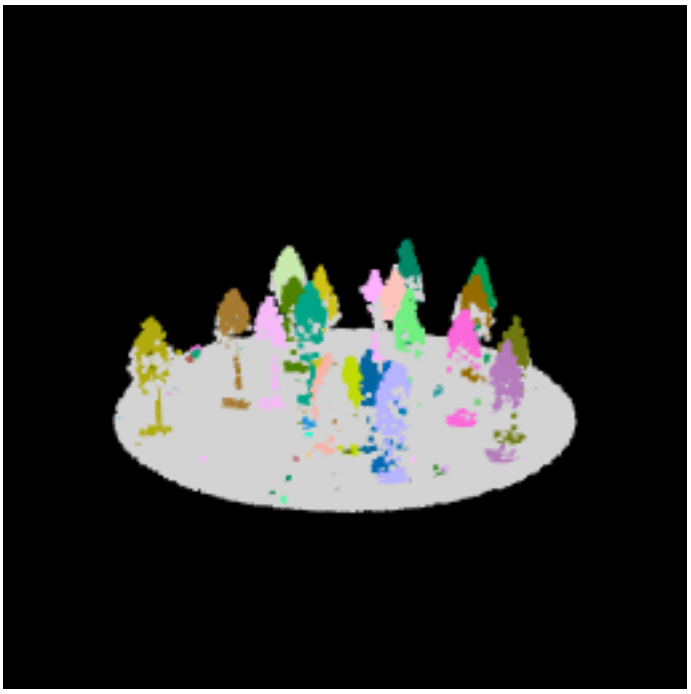
Dalponte Segmentation

Screenshot 2: Canopy Height Model of Plot 1 Overlaid with the Tree Tops

```
algo_dalp <- dalponte2016(chm = chm, treetops = ttops)

p1_dalp <- segment_trees(p1, algo_dalp)

rgl::close3d()
rgl::open3d(useNULL=TRUE)
```

```
## null
##    9
```

```
plot(p1_dalp, color="treeID")

rgl::rgl.snapshot("Outputs/Figure3_Plot1_Dalponte.png")
rgl::close3d()

knitr::include_graphics("Outputs/Figure3_Plot1_Dalponte.png")
```



```
p2 <- readLAS("Plots/Lab4_Plot_2.las")
```

```
p2 <- filter_poi(p2, Z >= 0 & Z <= 65)

res_list <- c(0.5, 2, 4, 10)
chms <- list()

for(r in res_list){

  max_edge_use <- rep(max(1, 3*r), 2) # scale max_edge with resolution

  cat("Building CHM at", r, "m with max_edge =", paste(max_edge_use, collapse=", "), "\n")

  chms[[as.character(r)]] <- rasterize_canopy(
    p2, res = r,
    algorithm = pitfree(
      thresholds = c(0,10,20,30),
      subcircle  = 0.2,
      max_edge   = max_edge_use
    )
  )
}
```

```
## Building CHM at 0.5 m with max_edge = 1.5, 1.5
## Building CHM at 2 m with max_edge = 6, 6
## Building CHM at 4 m with max_edge = 12, 12
## Building CHM at 10 m with max_edge = 30, 30
```

Screenshot 3: CHM of Plot 2 at 4 different resolutions (0.5m, 2m, 4m, 10m)

```
## Warning in plot.window(...): "zebra" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "zebra" is not a graphical parameter

## Warning in title(...): "zebra" is not a graphical parameter

## Warning in plot.window(...): "zebra" is not a graphical parameter

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## Warning in plot.xy(xy, type, ...): "zebra" is not a graphical parameter

## Warning in title(...): "zebra" is not a graphical parameter

## pdf
##   2
```
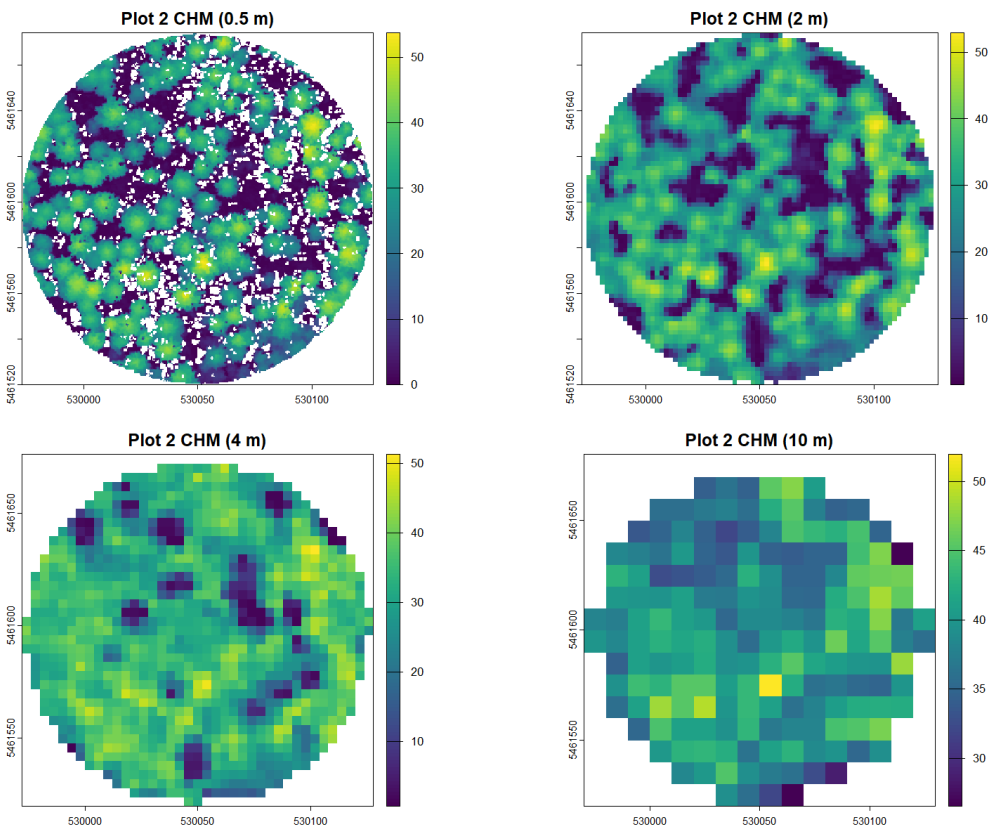
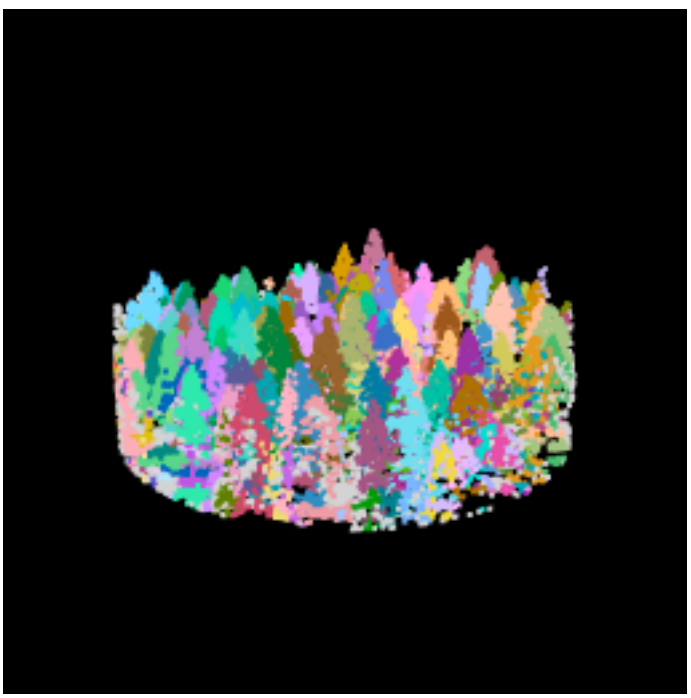



Screenshot 4: RGL Plot of Plot 2 at 2m Resolution

```
r_use <- 2
chm_use <- chms[[as.character(r_use)]]
tt2 <- locate_trees(p2, lmf(ws = 5, hmin = 2))
p2_dalp <- segment_trees(p2, dalponte2016(chm = chm_use, treetops = tt2))

rgl::close3d()
rgl::open3d(useNULL = TRUE)
```

```
## null
##   11
```

```
plot(p2_dalp, color = "treeID")

rgl::rgl.snapshot(paste0("Outputs/Figure5_Plot2_Dalponte_", r_use, "m.png"))

rgl::close3d()

knitr::include_graphics(paste0("Outputs/Figure5_Plot2_Dalponte_", r_use, "m.png"))
```



Question 6

If you had to select a spatial resolution to conduct segmentation at for the entire MKRF, what would it be? Justify your answer.

2 m. This resolution preserves the crown shape and maintains computational efficienc. The 0.5 m CHM captures fine structural variation but introduces excessive noise and over-segmentation. Conversely, the 4 m and 10 m resolutions oversmooth canopy structure and result in under-segmentation and loss of individual crown detail.